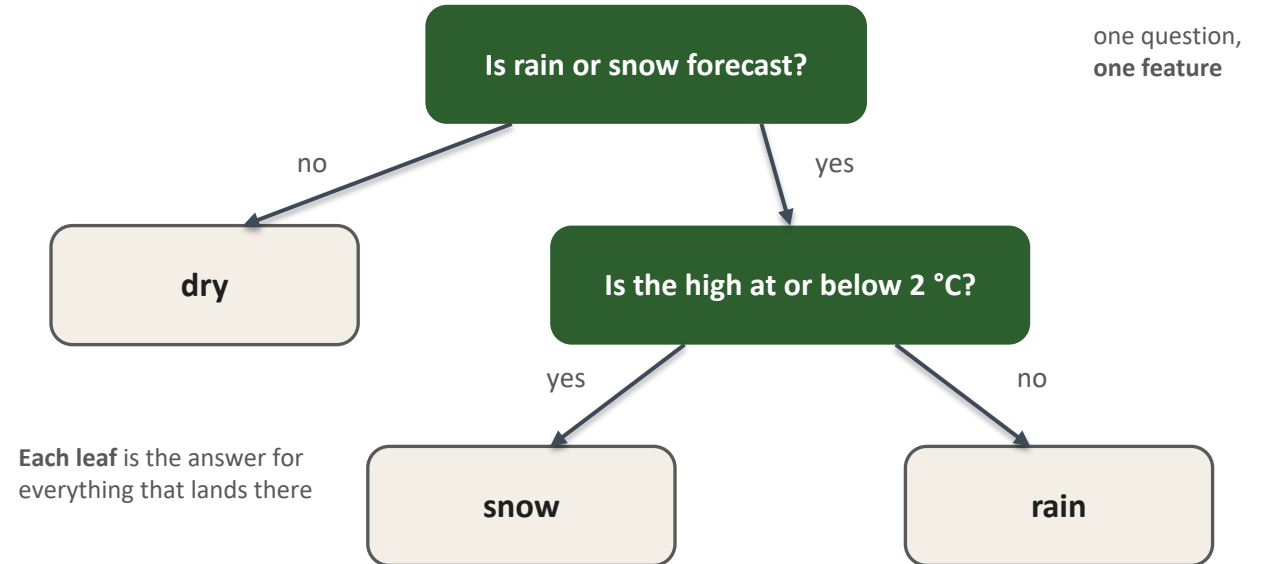


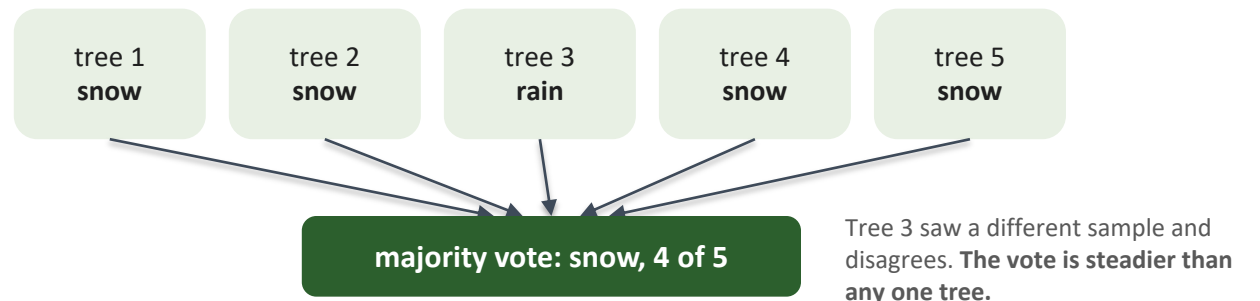
Decision trees and random forests

- **New model family.** A tree asks one yes-or-no question about one feature at a time and partitions the feature space into boxes
- **No equation, no scaling, no assumption about shape.** The tree learns the shape from the data. No comparison between features
- **A forest averages many trees** and is a workhorse for tabular environmental data
- **Today:** trees, overfitting, regression trees, forests, feature importance, and limitations

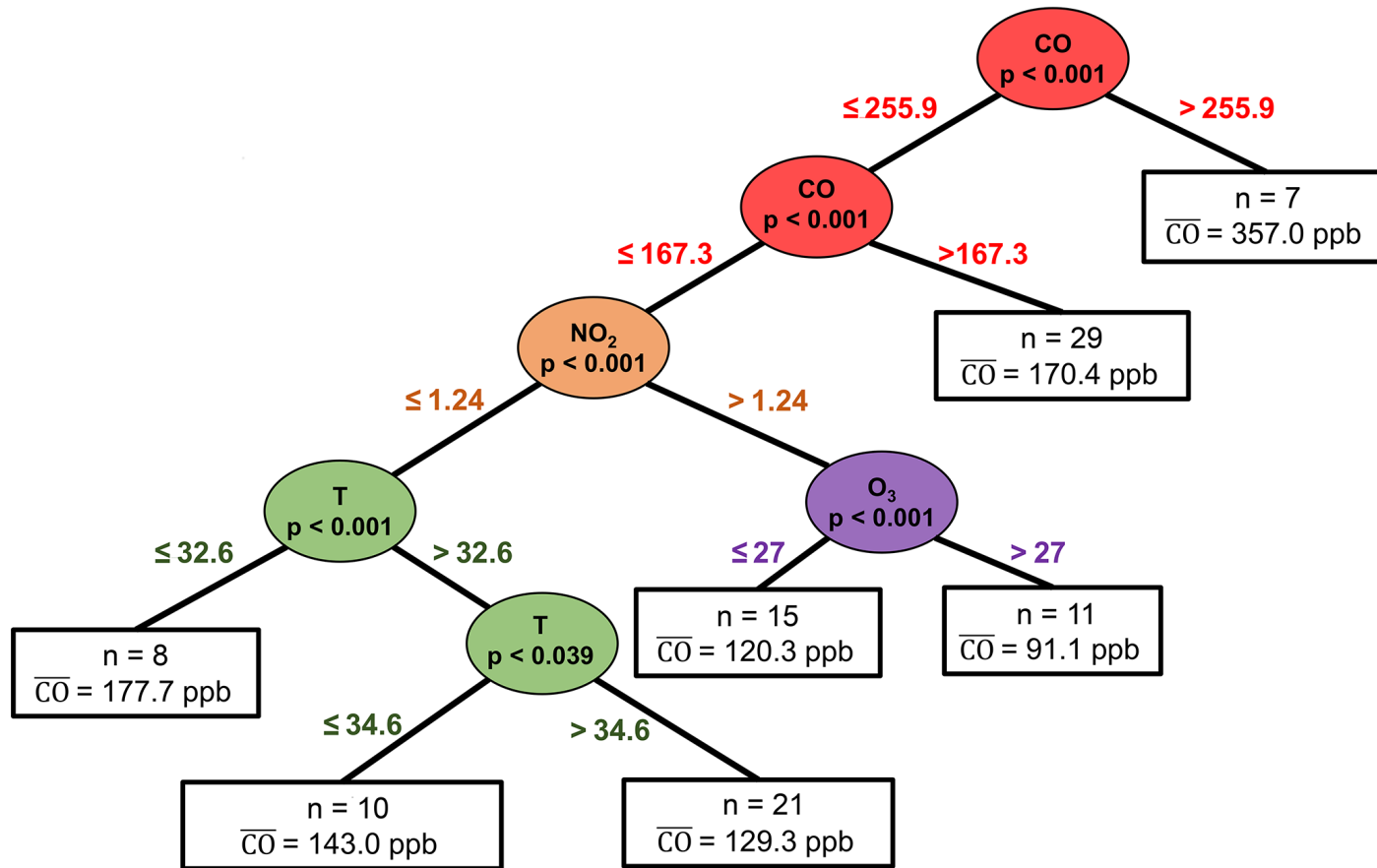
A **decision tree** is a flowchart of yes-or-no questions about the features



A **random forest** is many trees, each grown on a different random sample, voting



A decision tree is a flowchart



- **A real tree from a published model.** Correcting a cheap carbon monoxide sensor against a reference monitor: one of the 500 trees in the forest
- **Each node asks one question about one input:** the CO sensor's signal, then the NO2 signal, then temperature, then ozone, each against a threshold
- **Each leaf predicts:** the mean reference CO of the n training points that landed there, in ppb
- **The path is the model.** Low CO signal, moderate NO2, cool air: 177.7 ppb. The tree learned that the sensor reads differently in different conditions
- **No scaling needed.** A threshold on a column survives any rescaling of that column
- **Fully transparent.** The fitted model is a list of rules you can read and argue with

Figure 2 of Zimmerman et al. (2018), A machine learning calibration model using random forests to improve sensor performance for lower-cost air quality monitoring, Atmos. Meas. Tech., 11, 291–313, CC BY 4.0

Random forests in climate research

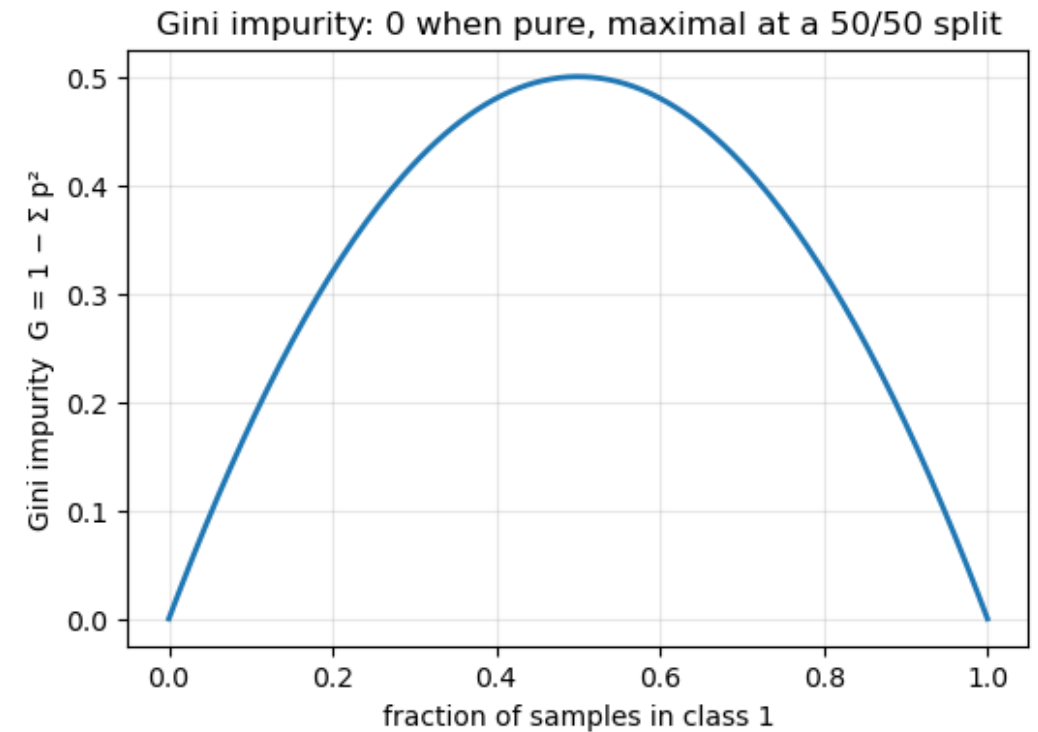
Problem	Predict	From	Type	Reference
Emulating an Earth system model: regional climate from emissions (ClimateBench)	Maps of temperature, diurnal range, rainfall, and extreme rainfall	Cumulative CO ₂ , CH ₄ , and aerosol emission maps, compressed with PCA	Regressor	Watson-Parris et al. 2022, J. Adv. Model. Earth Syst.
Calibrating low-cost air quality sensors	Reference-grade CO, NO ₂ , O ₃ , CO ₂	Raw sensor voltages, temperature, humidity	Regressor	Zimmerman et al. 2018, Atmos. Meas. Tech.
Global maps of carbon and energy fluxes from tower sites (FLUXCOM)	Net ecosystem CO ₂ exchange, latent and sensible heat	Satellite vegetation indices, meteorology, land cover	Regressor	Tramontana et al. 2016, Biogeosciences
Post-processing ensemble weather forecasts	Calibrated distribution of temperature or wind	Raw ensemble members and their spread	Quantile regression forest	Taillardat et al. 2016, Mon. Weather Rev.
Land cover and crop maps from satellite imagery	Class of each pixel	Multispectral bands, texture, time series of indices	Classifier	Belgiu and Drăguț 2016, ISPRS J. Photogramm. (review)
Streamflow, groundwater, and water quality	Flow, nitrate, recharge	Climate, catchment attributes, past observations	Both	Tyralis et al. 2019, Water (review)
Inferring the cause of a wildfire from its ignition point	Natural or human cause, then 12 specific causes	Weather, fuel, terrain, and social attributes where and when the fire started	Classifier	Pourmohamad et al. 2025, Earth's Future

The common pattern: a table of mixed inputs, a response that bends and interacts, a need to know which inputs mattered, and no interest in extrapolating past the data. That is the recipe you will run on electricity demand today.

References listed as first author, year, and journal. All are peer-reviewed applications or reviews of random forests.

How a split is chosen – classification trees

- **CART (Classification and Regression Trees):** at each node try every feature and every threshold, keep the split whose children are purest, then recurse
- The goal is to predict a class, so splits are chosen that separate classes as cleanly as possible (e.g. rain or snow)
- **Gini impurity of a node:** zero when pure, largest at an even mix. p_k is the fraction of the node's samples in class k
- **Cost of a split:** the children's impurity, each weighted by its share of the samples. CART picks the feature k and threshold t_k that make it smallest
- **For regression** swap Gini for squared error within the children



Gini impurity

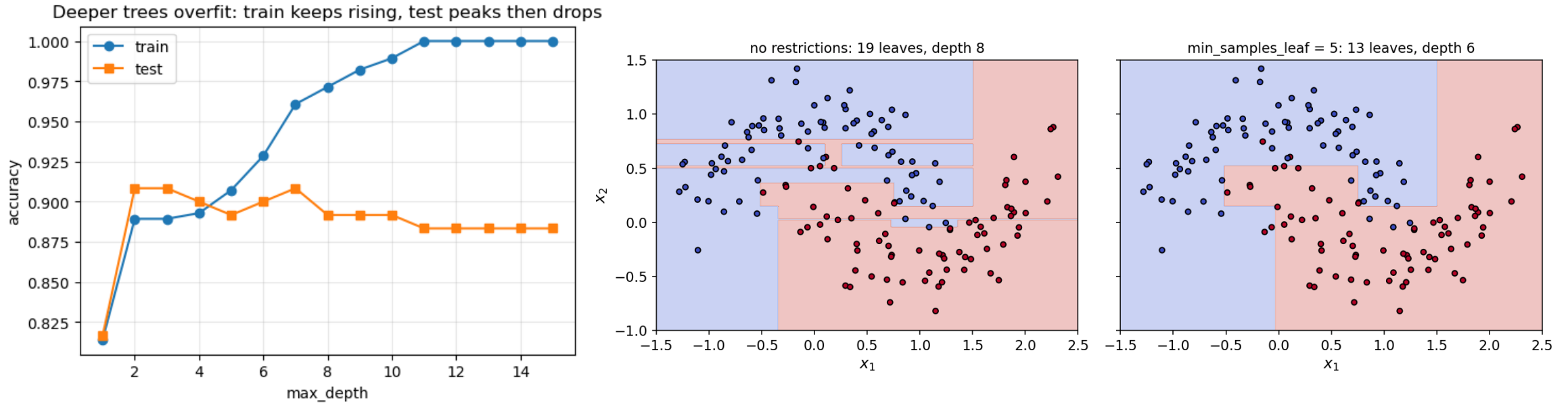
$$G = 1 - \sum_{k=1}^K p_k^2$$

cost of a split

$$J(k, t_k) = \frac{m_{\text{left}}}{m} G_{\text{left}} + \frac{m_{\text{right}}}{m} G_{\text{right}}$$

Figure from the Week 5 lecture notes; algorithm after Géron, Ch. 6

Overfitting, and the knobs that stop it



- **An unrestricted tree memorizes the training data**, noise included: training accuracy reaches 1.0 while test accuracy peaks and falls
- **The knobs limit how far a tree chases individual points:**
 - max_depth (caps number of questions),
 - min_samples_leaf, (minimum number of data points a single leaf must have)
 - min_samples_split, (only allow a split if node has at least this number of data points/samples)
 - max_leaf_nodes (cap on total number of leaves in the whole tree)
- **Choose their values on validation data**, never on the training curve, which always says "deeper"

Regression trees

- **Leaves predict the mean** of the training targets that reached them
- **Splits minimize squared error** within the children, instead of Gini impurity
- **The fit is piecewise constant:** a staircase with one step per leaf. Depth 2 gives four possible outputs, depth 3 gives eight
- **More steps where the curve is steep**, because that is where one mean fits worst
- **Same overfitting, same cure:** `min_samples_leaf` turns 200 leaves tracing noise into 15 that follow the curve
- **Every output is an average of training points.**

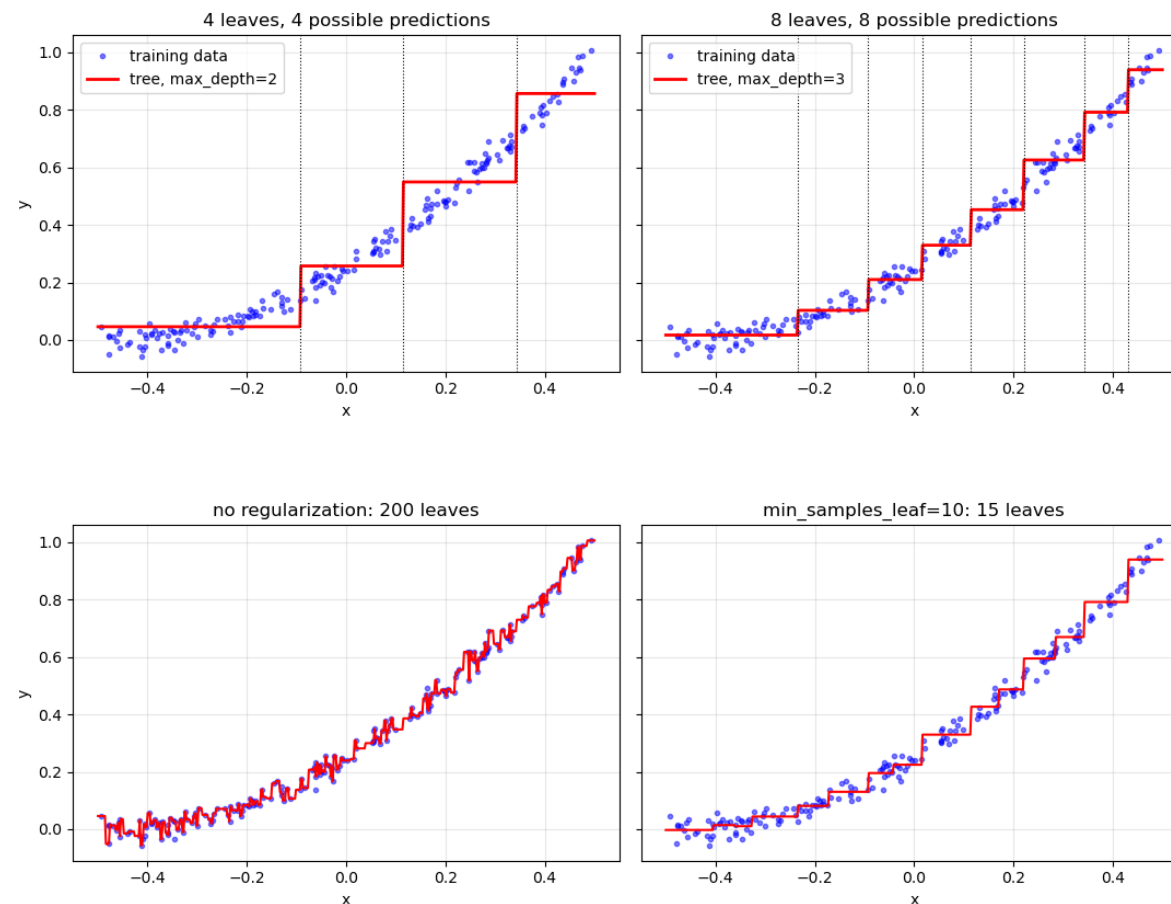
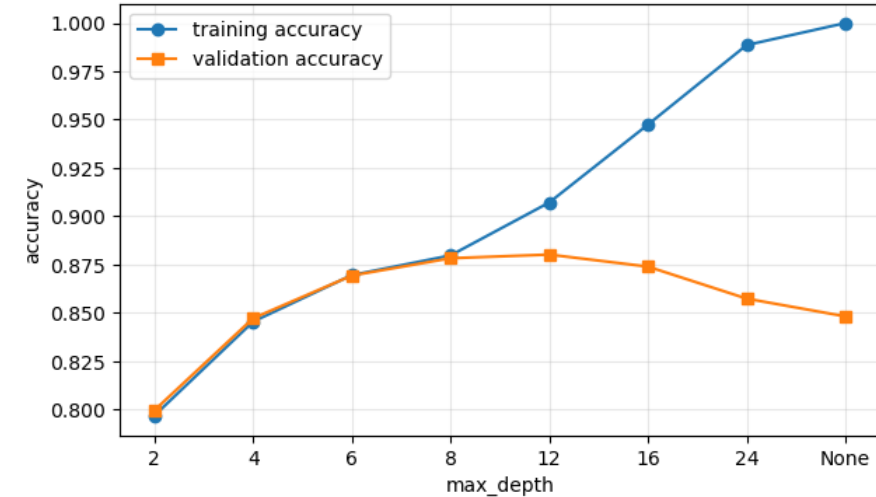
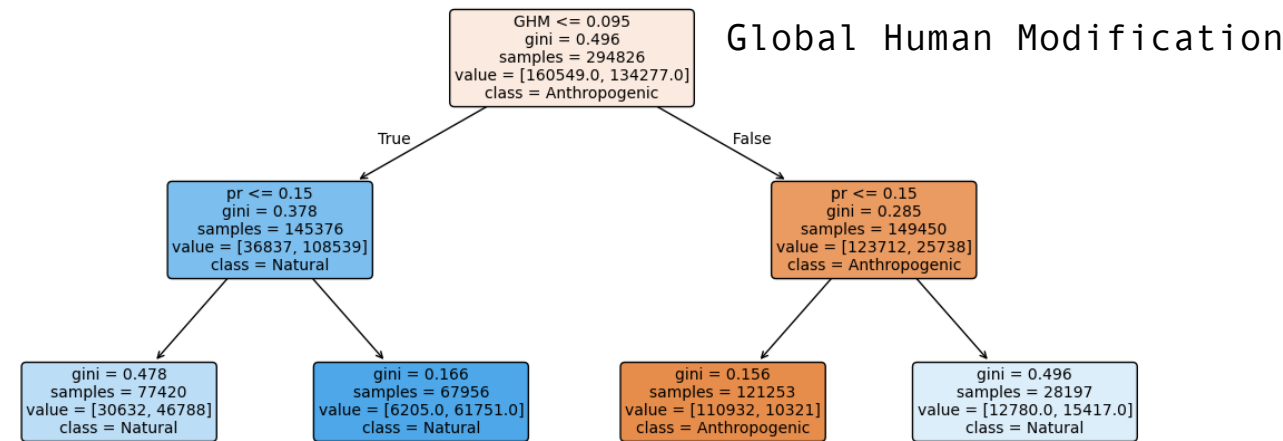


Figure from the Week 5 lecture notes, after Géron, Ch. 6

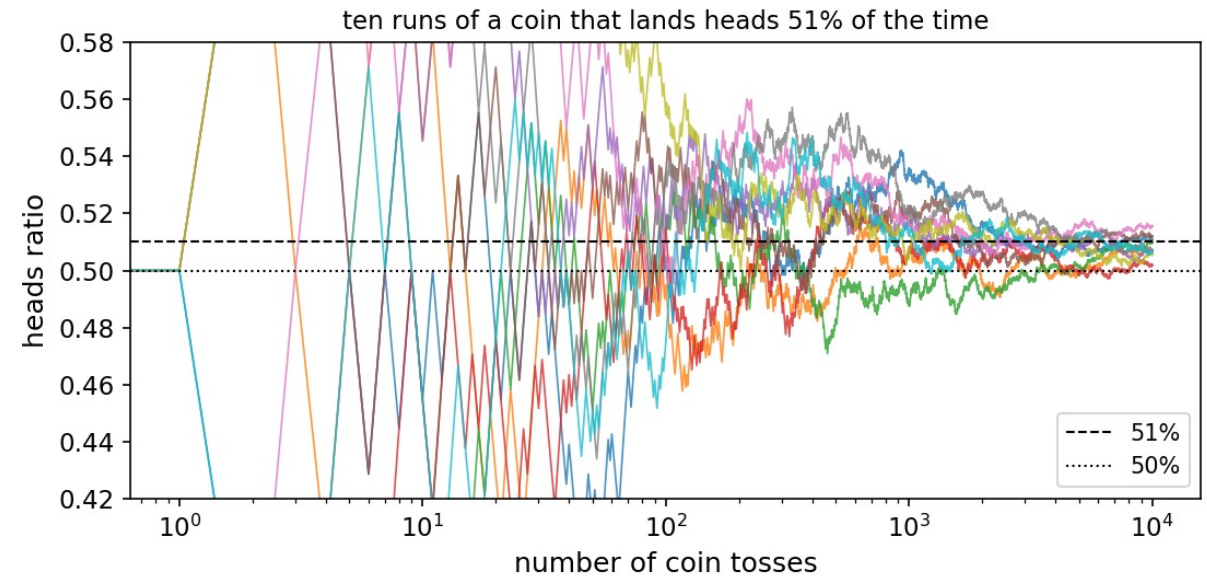
A real tree: what caused the fire?

- **Half a million western US wildfires**, each with weather, fuel, terrain, and social attributes at the ignition point. Natural or human-caused?
- **First question the tree found:** is the land within a kilometer barely modified by people? If so, lightning, right three times in four
- **Second question:** did rain fall that day? Thunderstorms bring both rain and lightning
- **Two questions, 80% accuracy**, and every step reads as physics
- **Depth sweep:** validation peaks at depth 12, training reaches 1.0.

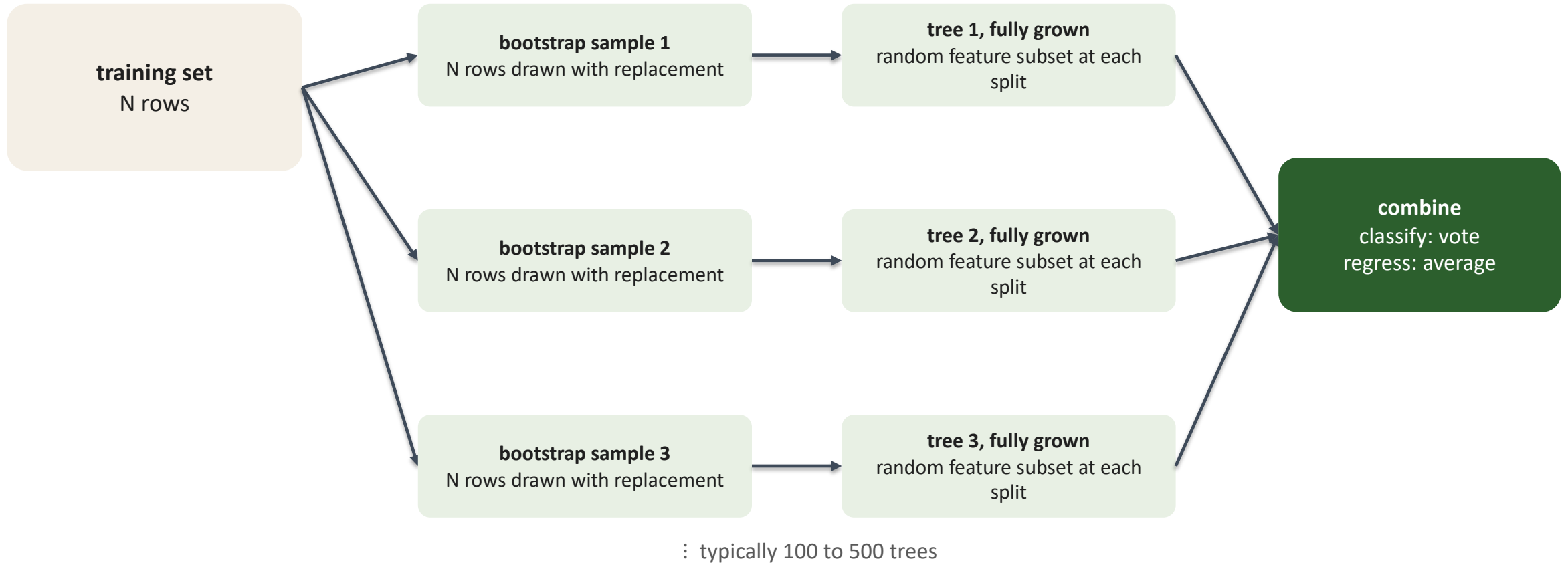


Why many mediocre models beat one good one

- **A coin that lands heads 51% of the time:** one toss is nearly a guess, ten thousand tosses land on 51% every time
- **A classifier that is right 51% of the time,** a thousand of them voting: the majority is right about 75% of the time
- **Only if their errors are independent.** A thousand copies of one classifier make the same mistakes and voting changes nothing
- **So an ensemble needs diversity,** members that are wrong about different things. The random forest is a recipe for manufacturing it

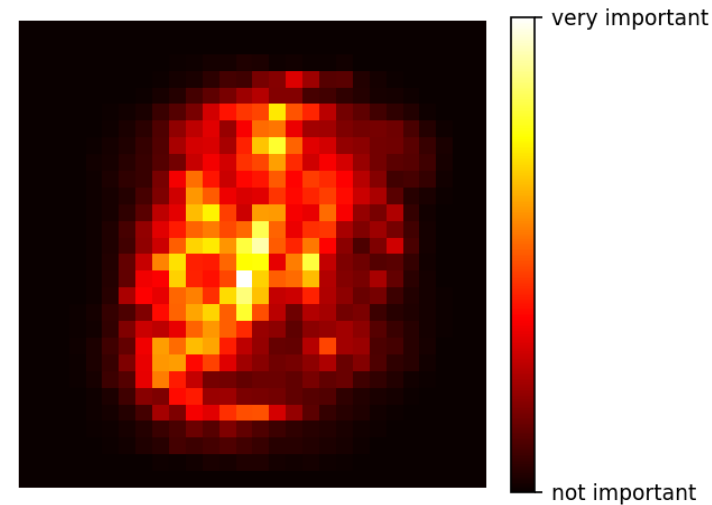
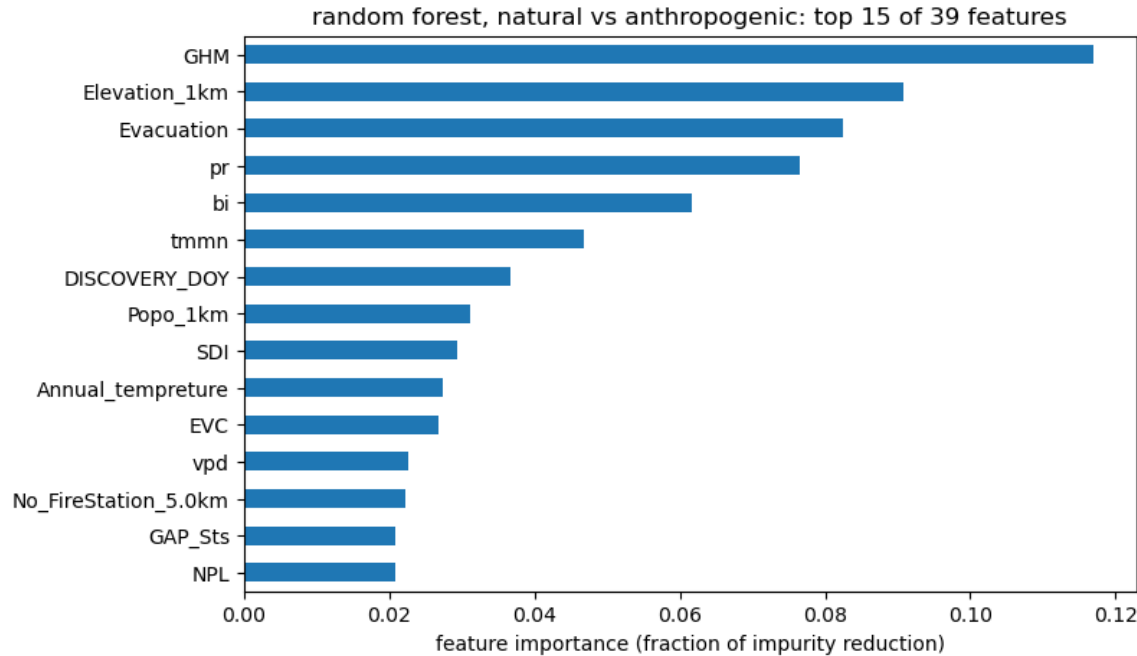


Bagging and the random forest



Bagging gives every tree a different sample. **Random feature subsets** make the trees disagree even more. **Soft voting** averages class probabilities. The rows a tree never saw score it for free: the **out-of-bag** estimate.

What the forest learned: feature importance

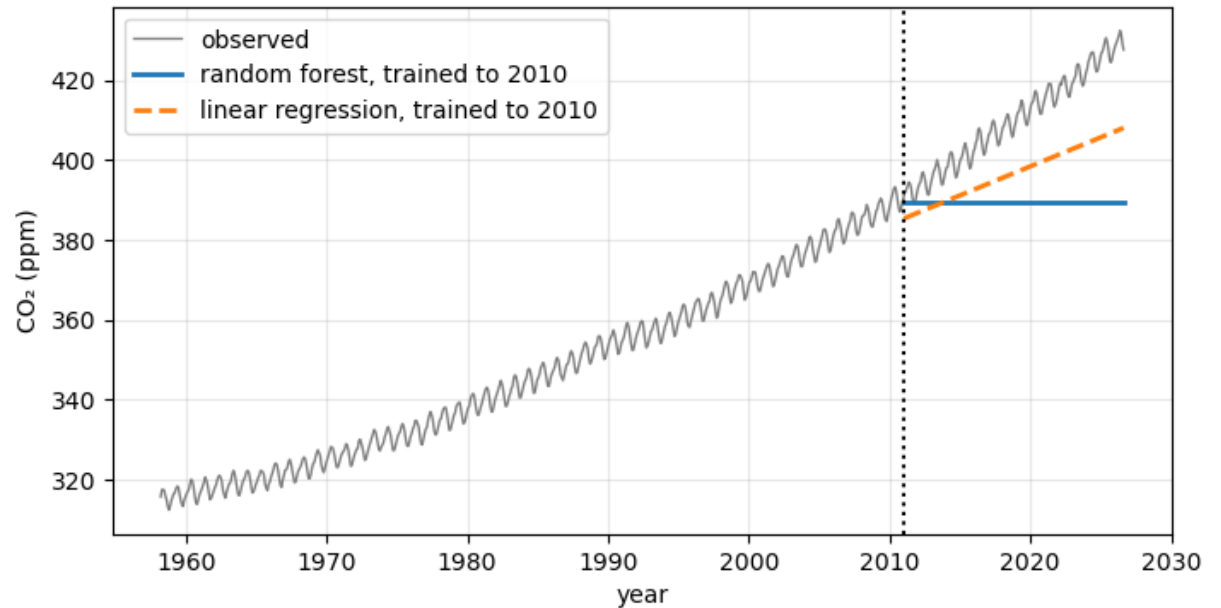
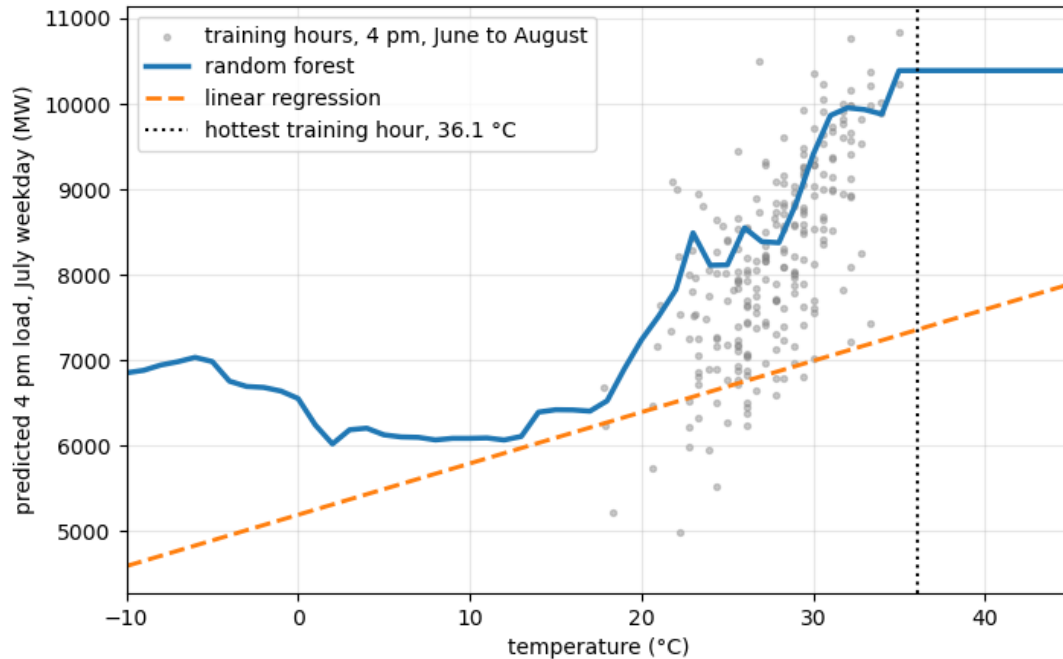


Géron's digits: the forest reads the middle of the image and ignores the edges.

- **For every split in every tree**, record which feature and how much it reduced impurity or squared error; sum per feature and normalize
- **Useful for splitting is not the same as causal.** And correlated features share importance, so a twin can hide a variable
- **Biased toward features with many distinct values.** For a scientific claim, confirm with permutation importance: shuffle one column, measure the drop

Left: week 5 tutorial on the wildfire cause data of Pourmohamad et al. (2025), *Earth's Future*, from Zenodo. Right: figure regenerated from the author's notebook for Géron, *Hands-On Machine Learning*, 3rd ed., Ch. 7, Ensemble Learning (Apache 2.0 code)

The one thing trees cannot do



- **A forest's prediction is an average of training targets.** It cannot leave their range, and it cannot continue a trend past the edge of the inputs
- **Past the hottest training hour the demand forest is flat:** the same load at 45 °C as at 36. On Mauna Loa, a forest trained to 2010 predicts 389 ppm forever
- **Questions outside the historical record,** a warmer climate, a record heat wave, get the edge of the data and no warning. Know which regime your question lives in

A practical guide

Use a forest when

The data are a table of mixed numeric and categorical columns

You expect interactions and nonlinear responses

You want a strong baseline with little preprocessing: no scaling, no encoding beyond integer codes

Rows are plentiful and prediction time matters more than training time

Prefer something else when

You must predict outside the range of the training data

You need a formula to interpret or to embed in a physical model

The inputs are images, text, or sequences, where networks win

Memory is tight: deep trees on millions of rows take gigabytes

Hyperparameters that matter

n_estimators 100 by default; more is better until it is only slower. **max_depth**, **min_samples_leaf** limit each tree; forests usually work with fully grown trees. **max_features** features considered per split, $\sqrt{n}$ for classification, all for regression. **class_weight="balanced"** for imbalanced classes. **n_jobs=-1** uses every core.

Cost: training scales as $\text{trees} \times \text{rows} \times \log(\text{rows})$; prediction is fast; memory scales with the trees.

Beyond bagging: boosting and stacking

Bagging

random forest

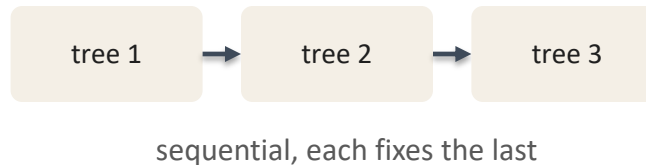
Trees grown in parallel on bootstrap samples; vote or average. One knob, hard to get wrong.



Boosting

AdaBoost, gradient boosting, XGBoost

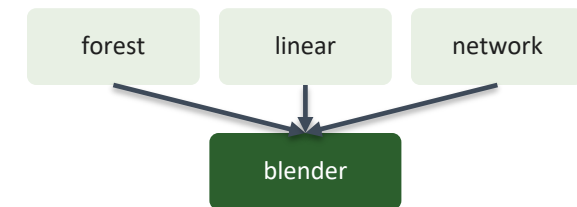
Trees grown in sequence, each fitted to the errors of the ensemble so far. Shallow trees, hundreds of them, a learning rate. Often the strongest on tabular data.



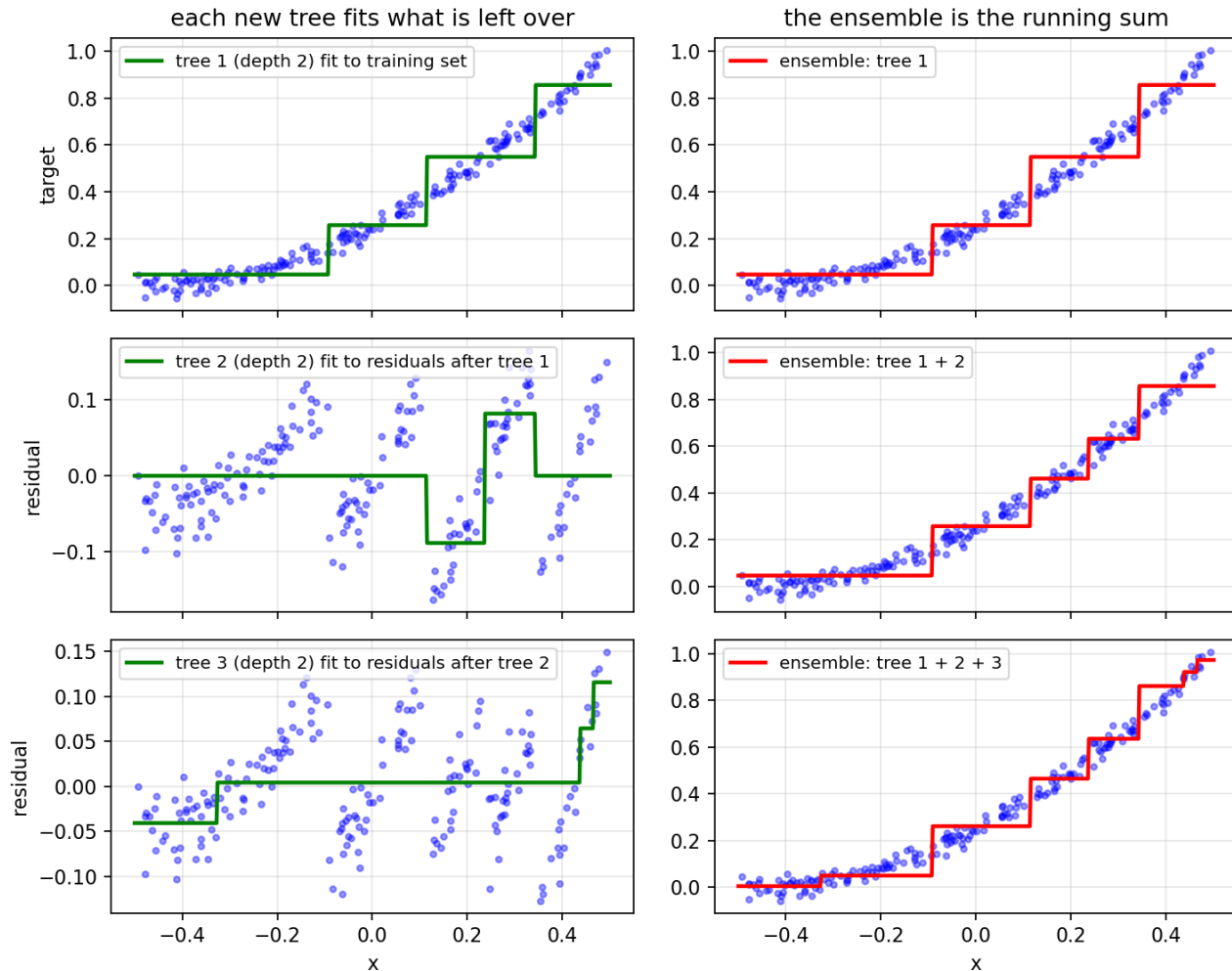
Stacking

blender over several models

Train different models, then train a final model to combine their predictions and learn when to trust each.



XGBoost: gradient boosting done carefully



- **Gradient boosting:** grow shallow trees one after another, each fitted to the residuals of the sum so far, and shrink each one by a learning rate
- **XGBoost** (Chen and Guestrin, 2016) is the fast, regularized version: a penalty on leaf count and leaf values, row and column subsampling, native missing values, parallel histogram splits
- **The knobs:** `learning_rate`, `n_estimators`, `max_depth` of 3 to 8, `subsample`. Pick `n_estimators` by early stopping on validation data
- **Same limits as a forest:** tabular data only, and it cannot extrapolate

New York City hourly demand	Validation RMSE	Test RMSE
Random forest, 100 trees	339 MW	346 MW
Gradient boosting, 300 trees, rate 0.1	321 MW	315 MW

Cousins: LightGBM and CatBoost are the same idea with different engineering, and scikit-learn's HistGradientBoostingRegressor is the same recipe without the install. **Cost:** trees are sequential, so training does not parallelize across trees, and there are more knobs to get wrong.

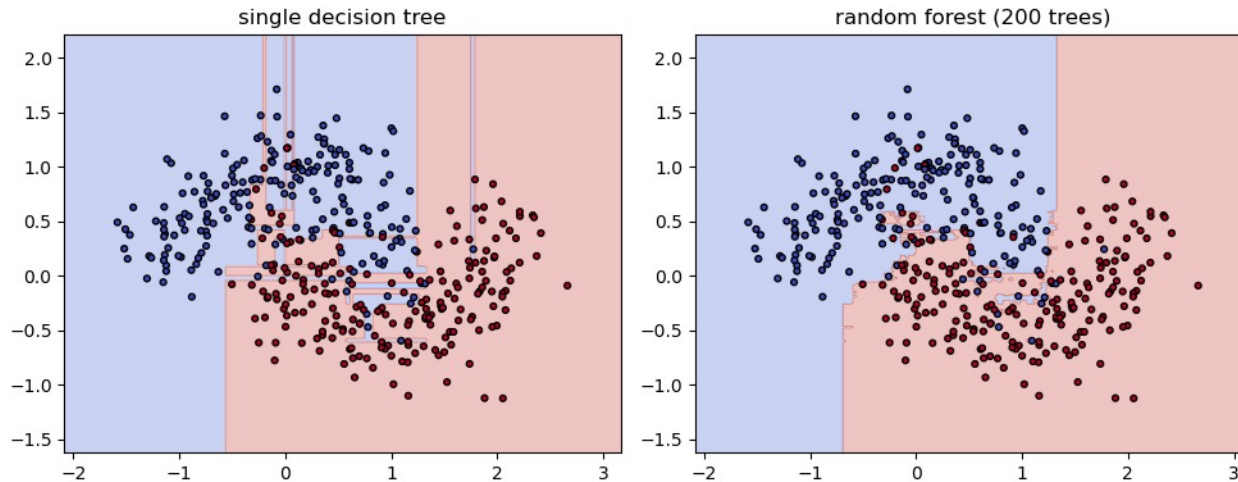
Left: boosting by hand on the Week 5 notes' quadratic, after Géron, Hands-On Machine Learning, 3rd ed., Ch. 7. Right: Week 5 tutorial data, scikit-learn's HistGradientBoostingRegressor

Summary

- **A decision tree** asks one threshold question at a time and partitions the feature space into boxes. It is transparent, needs no scaling, and overfits if left alone
- **Regression trees** predict the mean of each leaf, so the fit is a staircase; splits minimize squared error
- **Depth and leaf size are chosen on validation data.** The training curve always says "deeper"
- **A random forest** averages many fully grown trees on bootstrap samples with random feature subsets. The overfits differ and cancel; the result beats any pruned tree
- **Feature importance says which inputs mattered; partial dependence says how.** Neither is causal, and importance is biased toward busy features
- **Trees cannot extrapolate.** Outside the training range a forest returns the edge of what it has seen, with no warning. Questions about a changed climate need a model that can leave the data

Extra

Forest against one tree: classification



Wildfire, natural vs human	Validation accuracy
Best single tree, depth 12	0.880
Random forest, 100 trees	0.907
Forest on its training rows	1.000

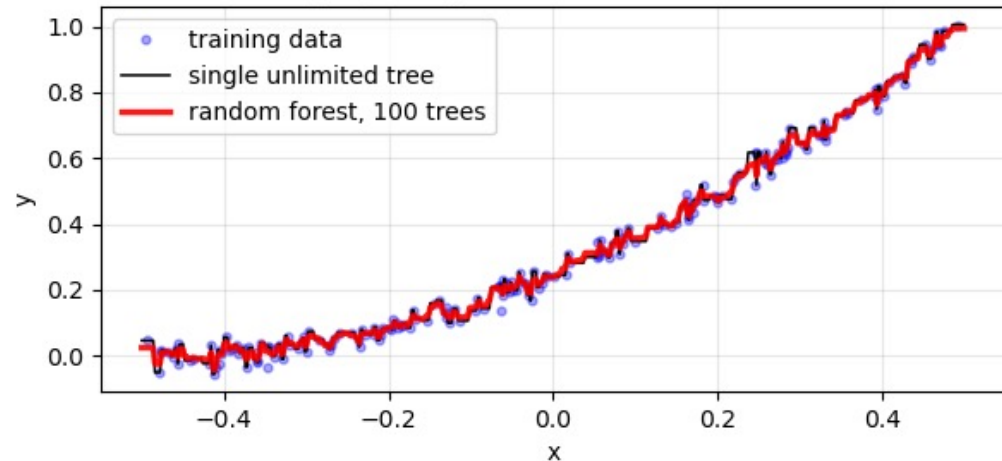
Trees in the forest	Validation accuracy
1	0.838
5	0.887
20	0.903
100	0.907
200	0.907

- **The forest's boundary follows the shape;** the single tree's follows the noise
- **Each tree scores 1.0 on its own sample,** yet the vote generalizes better than a pruned tree: the overfits are different and cancel
- **Five trees beat the best single tree;** past fifty, nothing changes but the bill

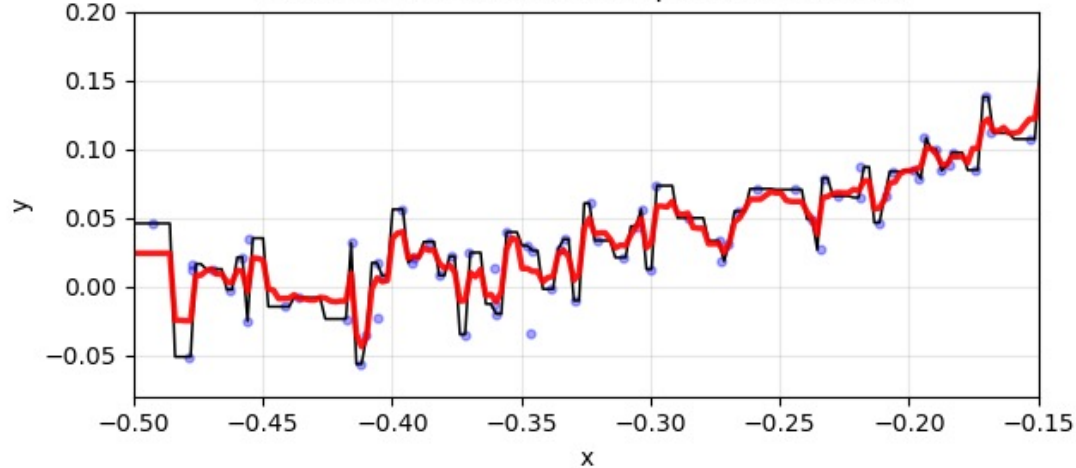
Left: figure from the week 5 lecture notes. Right: week 5 tutorial on the wildfire cause data of Pourmohamad et al. (2025), Earth's Future, from Zenodo

Forest against one tree: regression

Averaging many overfit trees smooths the fit



Zoom on the lower left: steps versus a curve



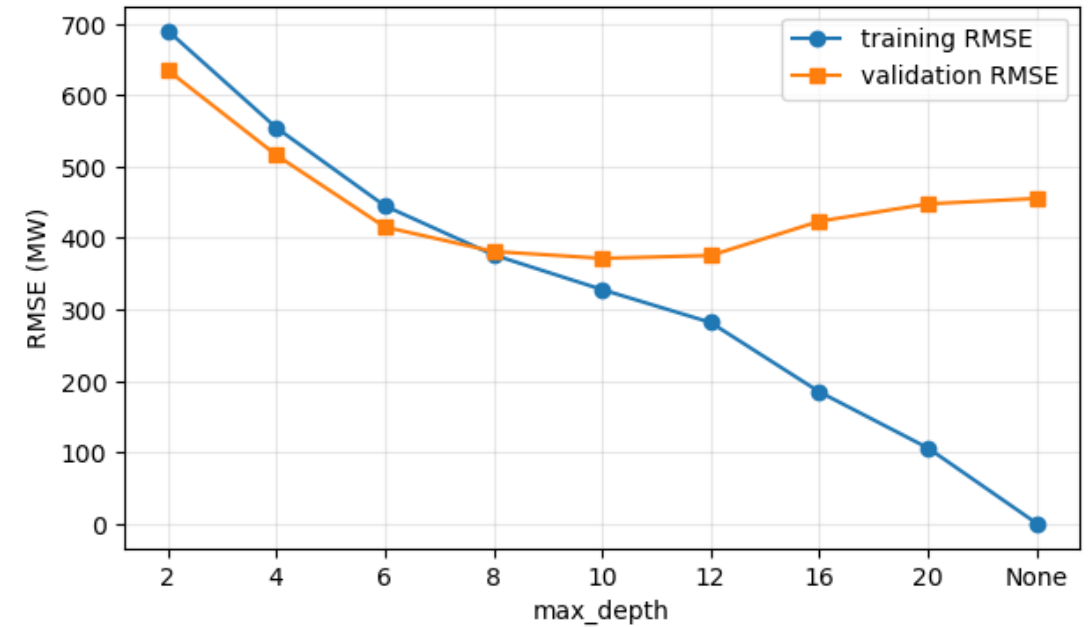
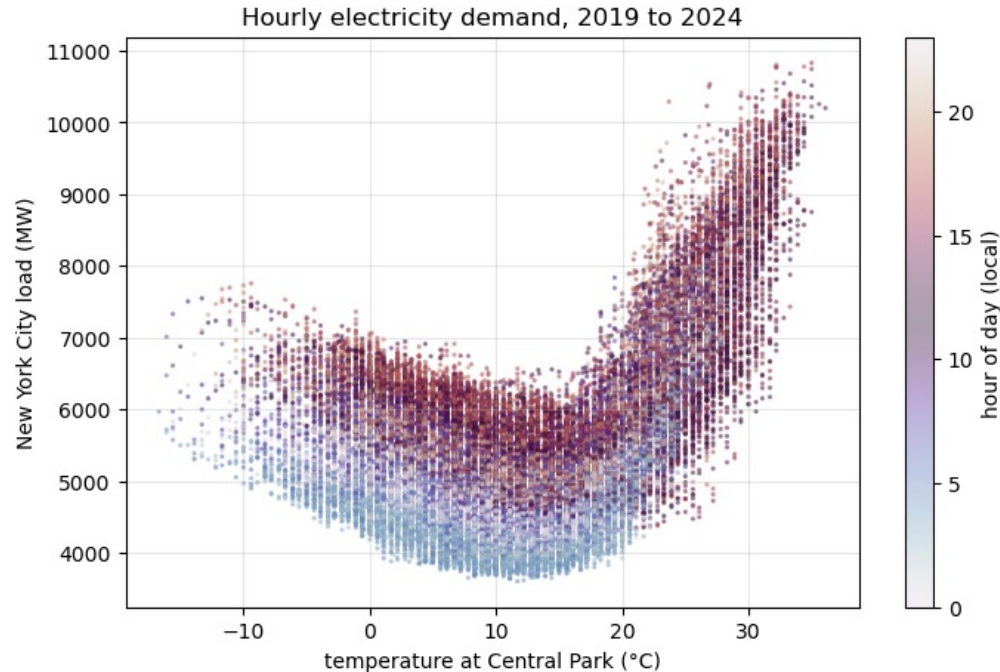
New York City hourly demand	Validation RMSE
Predict the training mean	1,109 MW
Linear regression (Week 4)	851 MW
Best single tree, depth 10	371 MW
Random forest, 100 trees	339 MW

Test RMSE on fresh points: single tree 0.035, forest 0.030, noise floor 0.025.

- **A regressor averages its trees' predictions** instead of voting
- **Many staircases averaged** make something far smoother than any one of them, close to the noise floor with no tuning
- **On real data the forest cuts the linear model's error by more than half.** The next slide shows why

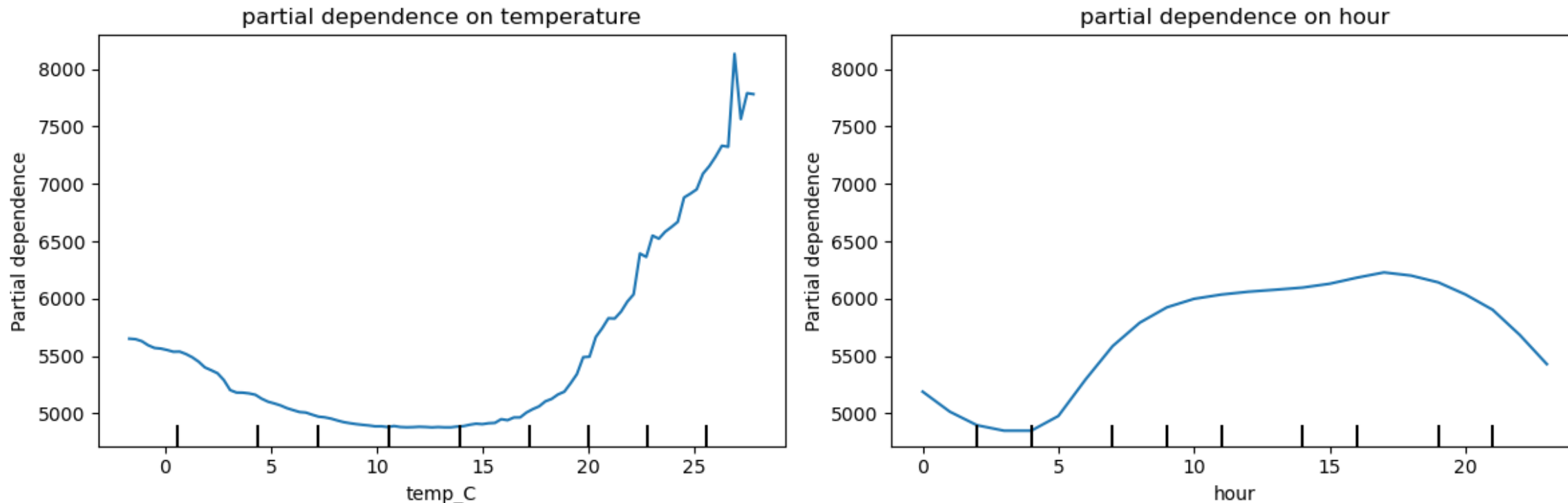
Left: figure from the week 5 lecture notes. Right: week 5 tutorial: NYISO hourly load for the New York City zone joined with NOAA hourly temperature at Central Park, 2019 to 2024

Electricity demand against temperature



- **A hockey stick:** demand falls as the day warms to about 15 °C, then climbs steeply as air conditioning switches on
- **A straight line through a U is the average of two opposite slopes.** The linear model says +60 MW per degree, everywhere
- **A tree does not care about shape.** It cuts at 22 °C, then 27 °C, then on the hour, and follows the curve

What the forest learned: partial dependence



- **Sweep one feature across its range**, set every row to that value, predict, and average: the model's response with the other features averaged out
- **The forest learned the U from the data.** Nobody told it demand bends at 15 °C. A linear coefficient answers the same question with one number
- **Caveat:** the sweep creates combinations that never occur, such as 40 °C in January, so read the ends of the curve with care